



Mastering Trigonometry: Heights & Distances!

The Ultimate Visual Cheat Sheet.



The Observer's POV



Angle of Depression
(looking DOWN)

Horizontal Level

Angle of Elevation
(looking UP)

 **TIP:** The angle is **ALWAYS** measured from the horizontal line! Never from the vertical wall!

The Trig-Picker Matrix

Use TAN



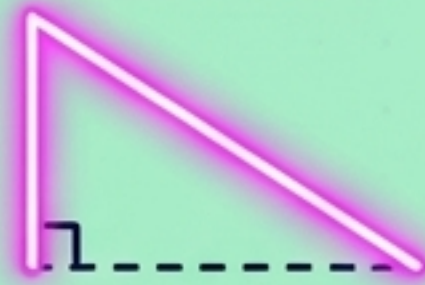
For distance from
base to height
(Opp/Adj).



Use SIN



For ladders, ropes,
and strings
(Opp/Hyp).



Use COS



For ground distance
when the hypotenuse
is known (Adj/Hyp).



The 4-Step Mastery Method

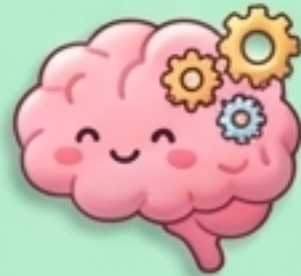
1. Draw & Label
(Don't panic!)



2. Spot the Right
Angle & find
Knowns/Unknowns



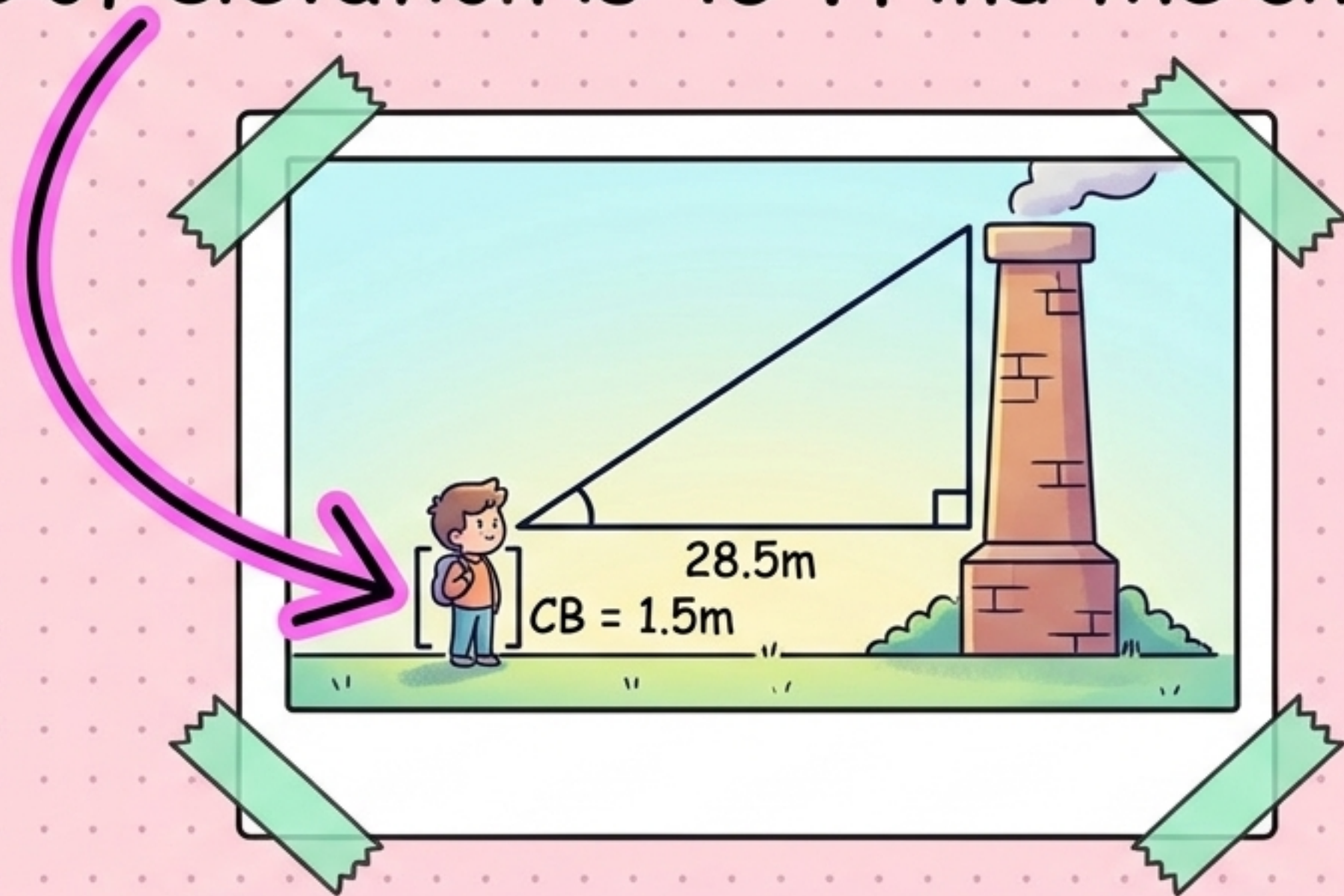
3. Pick your
Trig Ratio
(SOH CAH TOA)



4. Solve for X!



An observer 1.5m tall is 28.5m away from a chimney. The angle of elevation is 45° . Find the chimney height.



TRAP!
The triangle starts
at eye level, not
the ground!
Add the 1.5m back
at the very end!

The Math:
 $\tan(45^\circ) = AE / 28.5$
 $AE = 28.5\text{m}$
Total Height = $28.5 + 1.5 = 30\text{m}$

Boss Level: The Electrician

Step 1: Find the actual triangle height!



(5m total - 1.3m gap = 3.7m)

Step 2: Pick $\sin(60^\circ)$ because we need the ladder (Hypotenuse)!

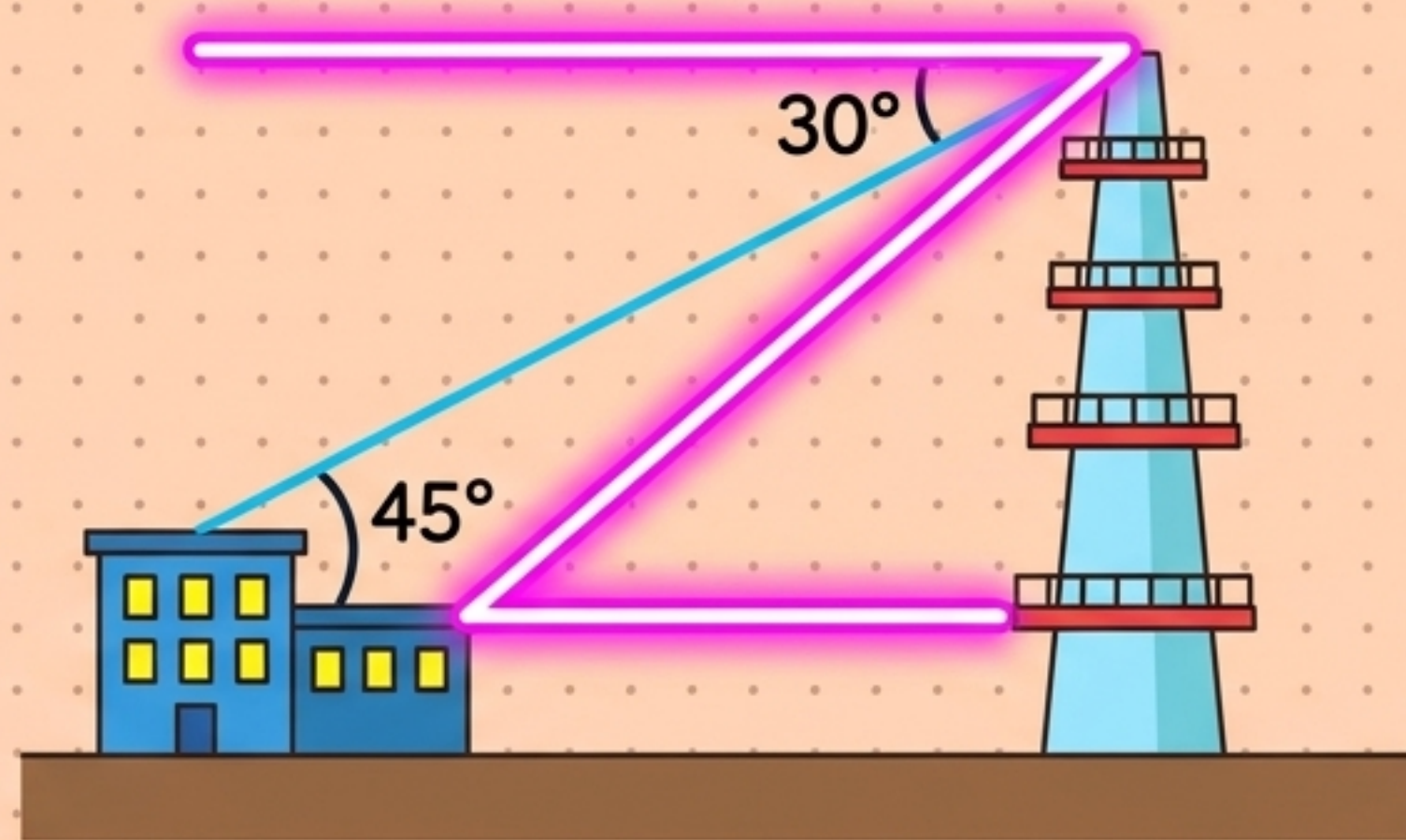
$$\sin(60^\circ) = \frac{3.7}{\text{Ladder}}$$

$$\left(\frac{\sqrt{3}}{2}\right) = \frac{3.7}{\text{Ladder}}$$

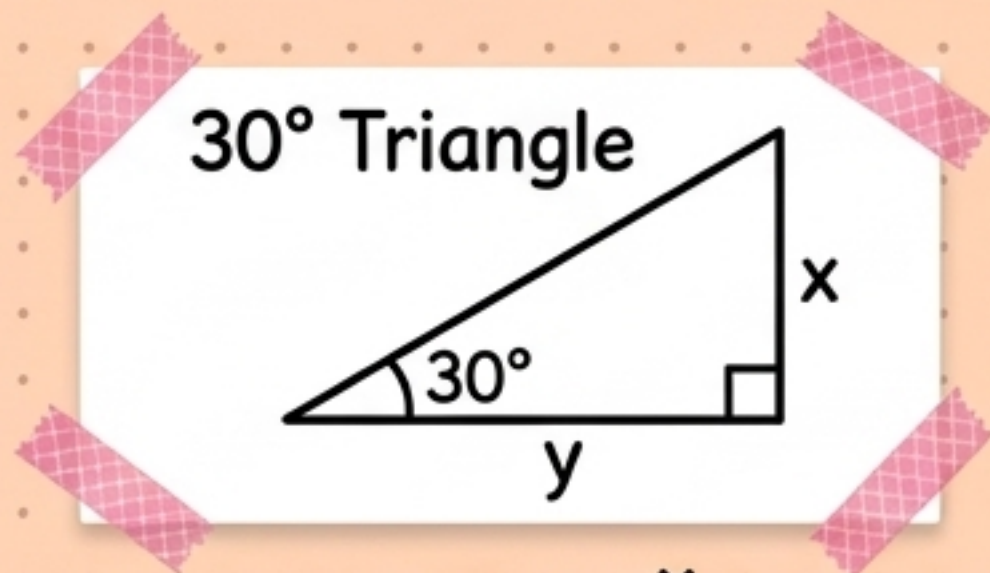
$$\text{Ladder} \approx 4.28\text{m}$$



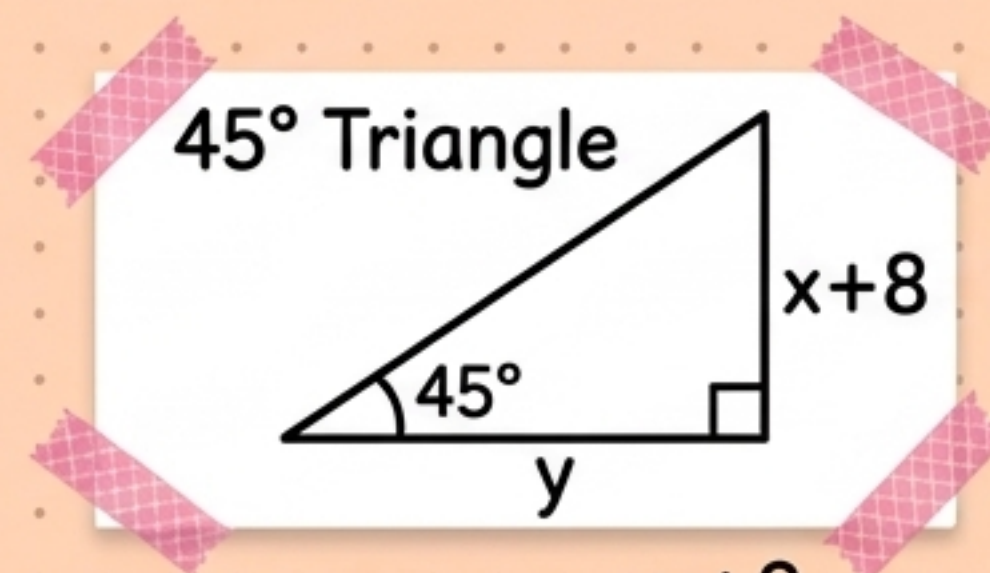
Boss Level: The Double Triangle



*Magic Trick:
Angle of Depression
= Alternate Interior
Angle!*



$$\tan(30^\circ) = \frac{x}{y}$$



$$\tan(45^\circ) = \frac{x+8}{y}$$

The Ultimate Cheat Sheet



Angle	Sin	Cos	Tan
30°	1/2	$\sqrt{3}/2$	$1/\sqrt{3}$
45°	$1/\sqrt{2}$	$1/\sqrt{2}$	1
60°	$\sqrt{3}/2$	1/2	$\sqrt{3}$

Quick Roots

$$\sqrt{3} \approx 1.732$$

$$\sqrt{2} \approx 1.414$$



Save this for your board exams! 💾❤️